

COMP.2560: System Programming

LAB #4: File Transfer / Coding

LAB 4: File Transfer / Coding

In this lab you will familiarize your selves with moving files between your laptop and CS Server, as well as

LAB 4: Overview

The main purpose of this lab is to:

- Part I: Familiarize yourselves with File Transferring.
- Part II: Modify code `"mycopy.c"`. **Submission is required.**
- Part III: Explain `"ex5.c"`. **Submission is required.**
See the description in Part III below for details.

These three parts of the lab are described below.

LAB 4 – Part I: File Transfer

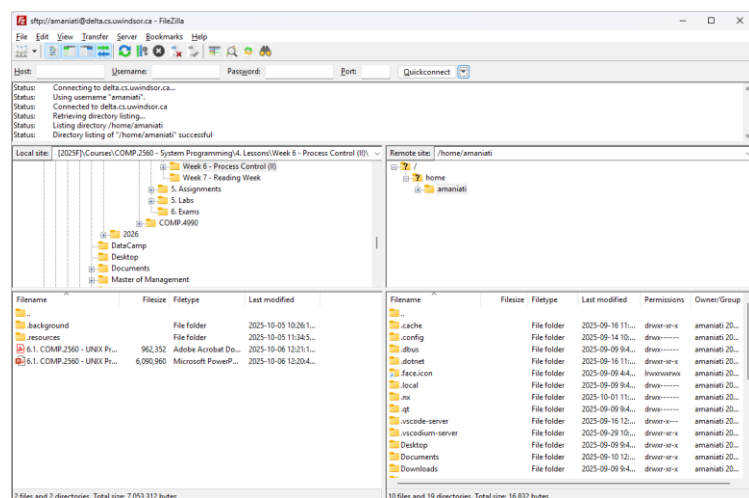
This part shows how you can transfer files between your local PC to and from the CS server.

Imagine you worked on the CS Server for your lab or assignment and finished it. It is now time to submit the required files to Brightspace and you face the problem of transferring the files from the CS server to your laptop and then submit.

There are at least two methods you can use:

- [SUGGESTED] Use FileZilla, or any other ftp client (for example, Bitvise).

Login to `delta.cs.uwindsor.ca` (CS server) using FileZilla. You should see the view presented in the screenshot below.



On the left-hand side, are folders and files on your local PC, and on the right-hand side, are directories (folders) and files on the CS server.

You can now navigate to any folder (directory) and copy/move/drag files between the CS server and your local PC.

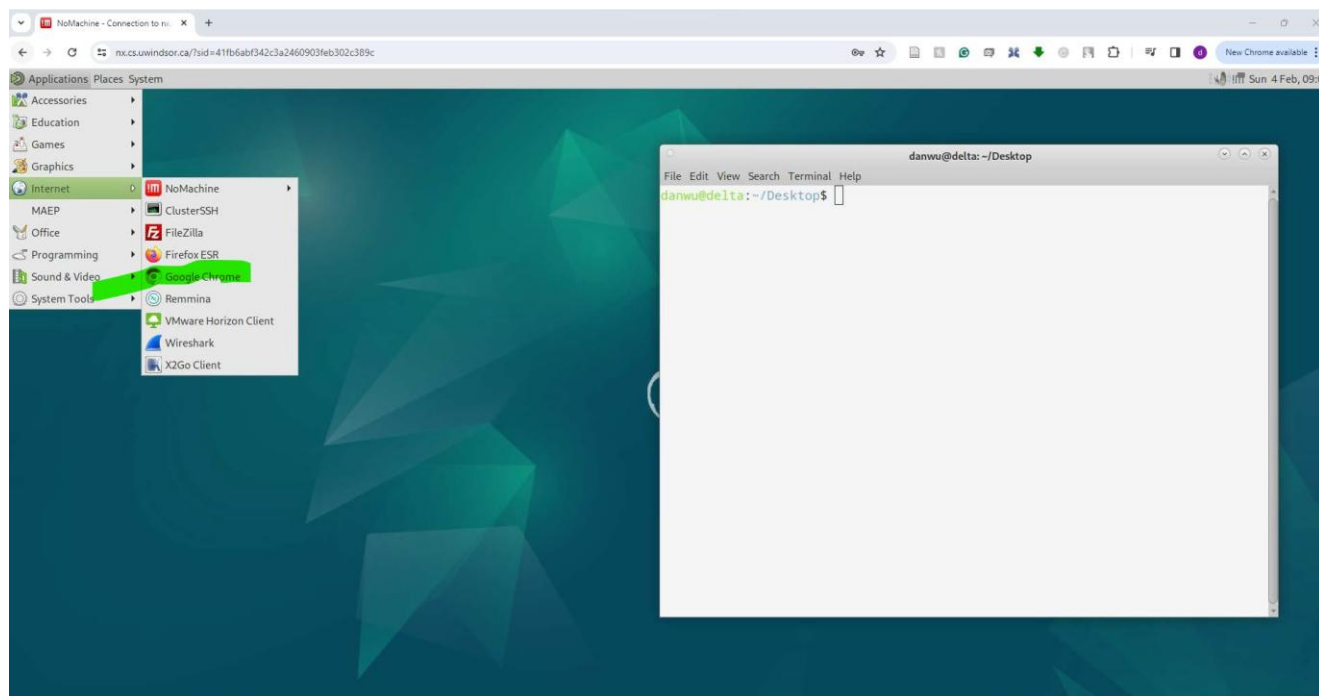
Once the files are on your local PC, you can then submit them.

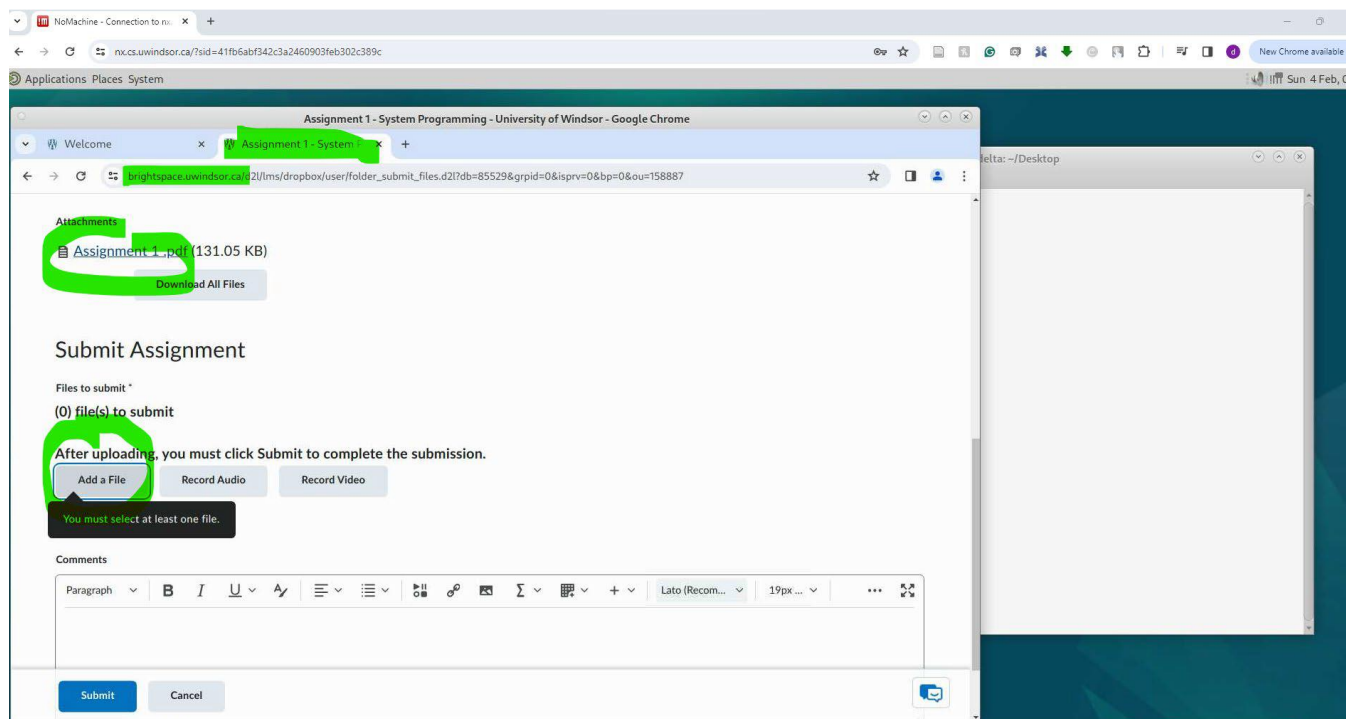
2. Use NoMachine from within a web browser or from the client application.

You previously experimented during Lab 1 on different methods to access the CS Server from within a web browser by connecting to <http://nx.cs.uwindsor.ca/>.

Once connected, you can launch the Chrome browser on the CS Server, and log in to your BrightSpace account, then navigate to the lab/assignment submission page for this course, and you can then choose the files to submit.

See the screenshots below:





No submission is required for Part I, it is for information purposes only.

LAB 4 – Part II: Modify a program

We mentioned quite a few times in some of the sample code discussed in class how to check the end-of-file (EOF) condition. We used a code segment similar to the one below:

```
while ((c=getc(inf) != EOF)
{
    putc(c, outf);
}
```

We emphasized that this is not the best way to check if we have reached the end of a file, unless you are sure 100% there are no read/write errors.

Do your research (Textbook, Chapter 5.6) and **modify the “mycopy.c”** program so that you have a proper check on the end-of-file condition. Put comments in your code.

Submit your source code and the script files (learned in Part I in Lab 2) at the end of your lab session. Please write down your lab section number in the submission.

Write a small text file to briefly explain what the problem is and submit it.

LAB 4 – Part III: Rewrite a program

We briefly talked about the “`ex5.c`” code in class and it has one problem. Download this code, study what it is supposed to do, and then run it:

1. Instead of pressing Ctrl+d (to generate EOF to exit the while loop), press Ctrl+c while running this program to abnormally terminate it to see what happens.
2. Write a short **txt file** to explain:
 - a. What this program is supposed to do, and
 - b. Describe your observations when pressing ctrl+d and ctrl+c, respectively.
 - c. What is the problem for this code when you press Ctrl+c?
3. Next study the `setvbuf (...)` function (Textbook, Chapter 5.4) and use the `setvbuf(...)` function to fix the problem in this code.

Submit the .txt file, source code of your modified `ex5.c` using the `setvbuf (...)` function, the script files (learned in Part I in Lab 2) running your modified `ex5` code.

LAB 4 – Submission

Submit the items elements described above.

To formally submit your work for this Lab:

- Submit a text file, briefly explaining what your findings are for Parts II and III.
- Submit the re-written code for `mycopy.c` and `ex5.c`.

When submitting, please indicate in the submission textbox the Lab section you belong to.

Evaluation:

1. It is our intention to evaluate your work being done in-person, while you are in the lab.
2. It is strongly suggested that you attend in-person to familiarize yourselves with the Lab infrastructure, meet with the GATAs and solve problems.
3. Each session is only 80 minutes long, so it is strongly suggested that you arrive early, are focused and do not disrupt the process.
4. It is desired that you complete your work while in the lab and get graded after live review from the GATAs.
5. Should you need more time, you can take your work home and present it to the GATAs on the very next lab session (the week following).
6. Even if you decide to work from home and submit your work remotely, you still need to have your work peer reviewed, in-person.

NOTE: Please remember that from this Lab 3 and on, **NO** video recording will be required, unless specifically stated differently. You need to submit only what is required by the lab. You **DO NEED** to have your work peer reviewed in the lab, **IN-PERSON**.